

ActInf Livestream #018.1 ~ The predictive global neuronal workspace: A formal act inf model

Source: [YouTube](#) **Playlist:** Active Inference Livestreams (Paper discussions)

Duration: 1:57:41 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome to the active inference lab this is the active inference lab and the active inference live stream today we are in active inference live stream 18.1 on march 23rd 2021 welcome to the active inference lab everyone we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at our links here this is a recorded in an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome here and we'll be using good video etiquette for live streams today we're in live stream 18.1 in our first of two participatory group discussions on the predictive global neuronal workspace paper and we're with the authors here ryan smith and christopher white so thanks to both of you for joining today today in 18.1 the goal is just going to be to discuss and learn and ask some questions about this awesome paper and we're going to go through some introductions and warm-ups and then i believe christopher will have some slides to share with us and set some context and then we'll just proceed with probably a bunch of questions that we all have about the paper as well as some questions from the live chat to get started though let's just go around and introduce ourselves and we can each just give a short introduction or check in and then pass it to somebody who hasn't spoken so i'm daniel i'm in california and i will pass it first to dean i'm dean i'm up here in calgary in canada and i'll pass to stephen hello i'm stephen i'm up in toronto in and i'll pass it over to christopher hi thanks i'm christopher white i'm a phd student at the university of cambridge and this is this page just to give some brief context this paper is really a work that kind of came out of my master's thesis that i was doing with ryan but we've kind of continued on um working on so hopefully we can chat about both this paper and then at some point chat about some things that we've been continuing on and so i'll talk past to blue hi i'm blue knight i am an independent research consultant based out of new mexico and i will pass it to dave dave douglas i'm in the mountains of the northern philippines where it's almost the end of spring um i'm retired from i.t uh especially from uh machine translation working on

the semi-technical aspects of our education project and then i believe ryan um yeah so i'm ryan smith um i'm an investigator at the laureate institute for brain research um and i yeah we're in tulsa oklahoma basically i run a lab focused on computational neuroscience and computational psychiatry well i'm sure we'll talk a lot more about that let's uh maybe just go right to christopher's slides and then we can pick up just with conversation or with this slide deck as needed but christopher go ahead and share your slides and i'll make them the full screen yeah sure okay okay so i'm assuming you can see my um screen are you saying we're seeing our jitsi uh okay how about now yep looks good okay well i now i it seems like i can't both look at the even of the dual monitor setup it doesn't seem like it it wants to share the same screen that jitsi is on and that's okay so it means i just can't see anyone um so just give me a shout um so this this is these are slides that uh from a talk i gave at ucl about midway through last year um and so i i may need some reminding of bits of this paper because we did write it quite a while ago although it did get only just get published in december last year okay so the idea with this paper was really just to kind of formalize some things that had been around in literature for a while and namely putting global workspace theory into kind of a predictive processing framework and so they've been so carfriston has a really nice 2012 paper that kind of did this somewhat formally uh jacob but it didn't really connect that deeply with the global workspace literature um at all really um jacob had in chapter 10 of his 2013 book yakupovi sorry um has a really wonderful chapter where he discusses the relationship between phenomenal unity and global workspace theory and how active and how predictive kind of the predictive processing framework in general and active inference kind of broadly conceived fit in with global workspace theory and how these ideas can kind of take help make sense of some puzzles within global workspace theory and also kind of give predictive processing a theory of consciousness as it were um and then in terms of i published a brief paper kind of building on jacob's work where i kind of said hey he is that's where the predictive global learner workspace comes from um or that term which is just a term that i kind of want to introduce just to denote basically kind of predictive processing approaches to a global workspace style architecture and so when ryan and i started chatting because of that paper we really our goal was to put some fairly vague idea conceptual ideas into a more computationally precise framework and that's what we did here um and so the idea was really just to use an active a fairly straightforward deep temporal uh partially observable markov decision process solved via active inference and show how by kind of tuning some of the parameters you can get basically all of the global work a very large perception of the global workspace literature and

all of those empirical results kind of just falling straight out um of the model and what's i think really important to me to my background is in cognitive science and philosophy um and so i did my undergrad in cognitive science and did my masters in a computational cognitive neuroscience lab um and so although i work on for a long time for a while i was really interested in visual awareness and visual consciousness i was around a lot of experimentalists and it's really important to me that all of these ideas are testable and we actually specify testable predictions and so really what we were doing in this paper was trying as best we could to one reproduce a bunch of existing results but also kind of extend the model and show where it actually gave testable predictions um and some of those have actually been confirmed um recently which is kind of nice so um i'll just kind of flip through some things there are just a couple things i want to highlight and so the first thing is just to kind of define terms uh so we called it um a active inference model of visual consciousness and that was really just a kind of so conscious is the way i think about at least maybe this is some idiosyncratic but confidence to me is just a kind of a catch-all term um it isn't one thing there are lots of sub things within consciousness that i think can come apart and so block kind of very famously in 1995 proposed the term access consciousness where this is just content that can be reported from the first person perspective and it is available to kind of cognitive control processes and working memory that's not a quote from him that's going to be paraphrasing but that's roughly speaking the idea and so that allows room for there to be other processes or unconscious processes saying the visual system of which you aren't aware and that can nonetheless influence behavior and he then also proposes it says chief in that paper his chief kind of worries distinguish between access consciousness and phenomenal consciousness where phenomenal consciousness is kind of the experiential aspect of this and he thinks that they so this is now and he thinks that they can come apart and that's what's controversial so i i don't personally agree that they can come apart but i i think it's a it's a useful distinction nonetheless and so i think in terms of thought experiment i don't remember where it might not be it might be in that paper it might be elsewhere but he's kind of talked about the example at least to me was really intuitive was if you imagine someone with a jackhammer or if you're doing if you're working at a desk and you can hear a jackhammer in the background you might what at some point you'll become aware of the jackhammer and but it was kind of a part of your experience all along you just weren't aware of it in a way um and to me that sounds rather odd but that's kind of but that's fine we can we can set that aside we're just defining terms here yeah i mean i mean obviously i agree i mean i think there's lots of cases like inattentional blindness and

change blindness and things like that where where you might you might think you were aware of something the whole time but work but anyway side thing yeah exactly i think like there are people i think i've forgotten who to coin this term but it's like the refrigerator light illusion like you open the refrigerator and the lights on do you think it was always on yeah it's kind of a nice analogy okay so just briefly about the global workspace so this idea kind of started in the late 80s with bernard bars and then stan haan's group really just took it and ran with it and so the idea is roughly speaking there's this prefrontal parietal network that connects in kind of motor processes attentional processing perceptual systems memory and all of these systems are in some sense competing for access and they are otherwise relatively kind of modular and disconnected but when they what is access to the global workspace or the content that kind of wins uh gains access to global workspaces broadcast throughout the system so it enables kind of otherwise isolate subsystems to share information in a common format and he's identified this with a prefrontal parietal and this like prima faces a bit of quite a bit of evidence for this so across meta analysis in report paradigms so actually people kind of like actually report their experience in binocular rivalry and phenomenal masking um there is a network of fronto parietal regions that is active for uh conscious but not unconscious you're cut off we can't hear you oh really i can actually still hear him oh maybe it was just me then oh good okay no worries thanks for pointing it out though ryan and uh yeah continue along girl yeah so there's basically the the long the short of it is a cross meta analysis that actually does seem to be quite quite a bit of evidence for the view that there's this frontal parietal network um and this story gets more complicated when we get into no report paradigms maybe we can chat about that at the end or next session because that's kind of the work that ryan and i are really kind of concentrating on the moment um okay i think we can so we'll i'll just kind of talk briefly about these two and then kind of move on to the model um so the idea so basically in a cross event related to um potential studies so this is a kind of contrast marking study on the right on the left and on the right there's an attentional blink i think goodness yes it is um basically what you can see is that there's this red line on the left and there is this little p3 it's like literally labeled with a p3b here and there is this late kind of seemingly non-linear all-or-nothing event-related potential over fronto central electrodes um that seems to characterize reported versus non-reported consciousness uh content sorry uh reporter bull rather not not um and one what this actually this is actually i should just flag for anyone who's kind of actually in the neuroscience of consciousness who is watching this this is no longer considered a neural coral of consciousness and i agree with that um it seems to drive be driven by task demands um but

nonetheless it is still a question why it is that when something is conscious and task relevant you get a p3b and that's kind of what we tackle in the paper and so really all of the evidence for the global work for global workspace theory i think it can be summed up quite nicely with this 2006 kind of taxonomy and so the idea is is that you can orthogonally manipulate stimulus strength and attention so similar strength you might just imagine that either might be like time of presentation so if you have a classic kind of masking experiment you'll have i don't know it might be a gabor patch then just after it there'll be some static noise and the static noise kind of masks is the mask to the gabor patch and depending on the type kind of the distance between the mask and the stimulus you can render something unreportable so and nonetheless it can still in some circumstances influence and so you can imagine that stimulus strength might be something like either you can you could get it might be contrast it might be the distance between the mask you could do both it doesn't really matter for our very coarse purposes here you can also vary attention so either have attention to the task or tension to be distracted away and the idea is that when there's a weak stimulus a bottom-up signal and attention is absent you basically just get very little activity it's whatever activity there is it's confined to very early striat cortex when you have attention is present but the signal is still weak you get some priming um but it's still unavailable for reports and so at that level starts to influence behavior then at the kind of the pre-conscious level or what they call pre-conscious borrowing a term from freud um this is where this is this is attention inattention blindness effectively so you have a very strong stimulus the gorilla can be walking across the room and wave at you it can be in the dead middle of the fovea and yet people still won't report it and the idea is the id this is kind of activity that's still very strong it's within the visual system but it's kind of gated out of um awareness via a lack of attention um at least most of the time and then when in this last quadrant here when you're attending something there's a strong signal you get this widespread kind of bifurcation in neural activity that characterizes aware versus unaware states and so we kind of had or personally i had a couple of reasons i was there a couple of reasons to be a bit suspicious about this and i actually do think web workspace theory has been updated to kind of accommodate these but nonetheless i'll go through them um so basically unconscious information can still be decoded uh from frontal regions so pfc activity is not synonymous with consciousness so the idea is that you could might have the idea that there is just a global workspace and you enter it associate loosely associated pfc and entering is sufficient for consciousness this can't be right um they have a solution to this they then basically introduced a bayesian model of conscious report where consciousness also requires a stimulus

representation to be separable from a noise distribution i really like this in the beginning of my master's thesis i spent a lot of time kind of playing around with the code it's really cool but i it's it's it's an ideal bayesian observer um it doesn't really give any neural predictions at all um i don't know and that that's that's that's fine limitations it's an ideal bayesian observer um that still would be nice if we had something a little bit more neurally detailed a christopher could you actually just unpack that slide 13 just what are the discs representing and how does this relate to bayesian statistics oh yeah sure so basically each of these axes is a source of evidence so evidence for stimulus y evidence for stimulus x and then um basically these two colored lines the pink colored line or the green colored line correspond to different decision criteria that you are imposing on a space so the idea is is that alternative to alternative first choice tasks where you're forced to choose between two things or versus report something as present or absent actually employ different decision criteria and so you can imagine these as a essentially a multivariate each of these as a gaussian distribution each of these circles as kind of a univariate gaussian distribution where if you are in the blue you that's stimulus to the left if you're all the way to the right or on x-axis that is stimulus on the right um where and then this green thing kind of close to the around the origin is a noise distribution so this is just saying there's nothing there and basically what conscious or awareness is on this account is basically these um distributions or either left or right being far enough away from the origin that they don't overlap substantially with the noise distribution so when you impose the decision criteria on the space you are confident that you have seen does that kind of make sense nice thank you cool okay um and then seeing this is unseen ignore ignores the noise distribution it's just separates left versus right so really cool model i really like it um but yeah unfortunately not quite as nearly detailed as i would like second problem is the p3b does not equal awareness so these are two paradigms so it's now being replicated in a no report condition basically a person is there's really sophisticated experimental techniques that allow you to basically access or gain access to a person make inferences about what a person saw without having to make explicit reports and when that happens you see that basically the p3b which was at one point taken to be a signature of just vanishes and that's a bit of a problem so we'd like an explanation of that and then the last one isn't really a problem so much as just a a point of absence so there's a really large body of evidence showing that expectation seems to be a pretty crucial factor in determining the content the visual consciousness so across paradigms continuous flash depression binocular rivalry motion induced blindness blah blah blah or there's all these areas where to show that expectations probably along the lines of

similar to attention should be a part of any taxonomy or any taxonomy that claims to kind of try to be complete of the factors underlying conscious access and the global neural workspace as it stands doesn't really describe how expectations are implemented and so our job or what we saw ourselves as doing really in this paper was just to kind of show how we could account for this existing work namely the taxonomy and also explain how the p3b doesn't equal awareness um or show hapso show how that dissociation arises very naturally and three show how expectations could be plausibly be implemented and give kind of a more give a model that generates predictions about expectations i should say and has a story about how they're implemented okay um and i think the the real thing is it's it's okay to revise a model like revising models is that that's just good science um and i think one of the part of global workspace theory that made me a little bit suspicious lately is that the revisions of problem one and two they didn't generate surplus neural predictions really they were just kind of they abandoned the p3b signature of conscious access and they event and they endorsed an ideal bayesian observer model of subjective reports but neither of those things they were kind of responses to problems that didn't themselves generate new predictions that go on um and that's kind of a characteristic of what's been called degenerative research programs by lactose and so i mean that we can argue about whether this is a fair characterization of global workspace theory but i think this is kind of at least a nice way of setting up where we were coming from okay um would you like to give a a primer on active inference for partial observable markov decision processes i'm not sure how much more you want me to talk or whatever um i mean i this was an hour long this is an hour i have an hour worth of um so i could talk forever but maybe you want me to skip to the results or just summarize the model or what let's actually uh hear your take here this is just a really awesome presentation and then everybody here maybe let's write down a question or two so then we can have another however many minutes you want of the slides have about an hour or a little more for questions and then we'll have next week for questions but it's really great to get this information out there because the way you're sharing it is awesome yeah cool okay well i'll i'll try and be a little bit more detailed when i go through the slides because i've kind of just been clicking through things at the moment so what just a brief this is this is a little bit of a point of confusion and ryan may want to kind of come up describe this as well so what is active inference well it's both a noun and a verb there's a noun that's you can use it to describe two things uh it's a it's just a mathematical framework that allows one to formulate models of kind scented behavior using equations that kind of justified a priori from a very general set of

assumptions about the nature of self-organizing systems it's also that you can attach a process theory to that mathematical framework that allows you to give predictions about various neurophysiological variables like firing rate of neuronal populations erps and kind of how neuromodulators like dopamine or noradrenaline affect things like firing rate in the rps and also behavior um it's also used as a verb and especially in kind of maybe pre-2016 literature that's basically what it was it was basically an extension of predictive coding to motor control and so people often use active inference just as a catch-all for basically the predicted coding version of motor control and there's been all sorts of confusion in the literature where people are taking kind of active inference as a theory basically of of markov decision processes right and they're conflating that which and because markov decision processes being a decision process is kind of a model of bayesian decision based optimal decision making and then if that's conflated with a theory of motor control or you use a motor control account of decision making you're gonna get say bizarre things um don't actually track what's going on yeah i mean i would that would that's that's the main point if anything that i would want to you know kind of come you know fully emphasize is that like old active inference is a theory of motor control i think they're old right i mean people still do it but yeah inference where that just means you know like hold a prior constant and make a reflex arc move the body to you know minimize the prediction error with respect to that prior about say like you know like proprioceptive position or something like that right that's just a theory about once you've decided what to do how to get the body to move accordingly it doesn't have anything to do with deciding what to do right so another control thing is just you know how do i move how do i get the body to move how i want to using something predictive coding ish right whereas you know we're talking about kind of like current is is again it's a it's a theory of um which is like it's totally different i really think these these these there should be two different terms right for these for these things yeah that's the conflation yeah exactly thanks and stephen do you have a quick question um yeah i'll just be curious like as you're here you know you're doing work in this paper here with the neuronal context of how deductive choices are made and it taps to the phenomenological questions as well do you think that's when these you see these problems happen a lot because when when people are you're trying to move between different aspects of um fields of practice which are sometimes thought about separately you know phenomenology phenomenology is often talked about in the inactivist field and you've got the neuroscience and they're often clean and separate but when you need to bring them together and try and talk about this bigger picture the distinction your the distinctions you're

making become even more important if that makes sense because there seems to be a lot of conflation that happens with that so i'll just be interested in your thoughts um so i think one thing to say is so i i like an activism um especially what's kind of been called computational activism i've forgotten there was a really awesome paper out in synthesis um i've forgotten the author name name of the author but sorry about that um but to me that stuff doesn't really have that much to do with what i would call consciousness or what i think of as consciousness as like a cognitive scientist i suppose um when i think of awareness i think of a very sharp my explanatory the thing that i'm trying to explain is very specific it is there are these states of which you are unaware some of which you can become aware of say for example like the intentional blindness case um or in phenomenal masking in some cases there is a visual stimulus in one case where it's unaware and there's a very small change physical change the stimulus but there's this radical non-linear change in whether you can report it or not i'm interested in that contrast between reportable stimuli and non-reportable stimuli now along with that comes when stimuli are reportable generally speaking there is a we might also be interested in characterizing kind of the content of experience as it were not just the availability for report um so we're going to get to that actually kind of towards the end of the presentation we actually do endorse a version of the phenomenal consciousness access consciousness distinction but really i think to answer just kind of repeat repeat myself somewhat and answer the question all that i'm trying to explain with this model is that distinction between content that's accessible and content that's not accessible everything else i don't really mind like people can call it whatever they like uh i'm not interested in modeling it yeah so i mean i mean in general right i mean again this just speaks to kind of having precise theoretical constructs and having a good kind of taxonomic taxonomy of these things within a particular area and also you know if you're going to try to do any kind of um interdisciplinary work right where you have constructs in one kind of field and you want to find some way to map them on to right or translate them into what you think they ought to be right in some other uh you know in some other theoretical language right then i mean then also specifying that in a precise way and how you kind of recover one from the other in terms of the way they're operationalized empirically right i mean it's also something that's that you have to do right to do this work well you know so i mean like you know chris already started right to kind of just people really loosely you know at least not in all cases right obviously some people are very precise but in some cases the word consciousness or awareness gets kind of thrown around to mean a ton of different things which makes it confusing and also makes the term you know uh

imprecise and therefore not really all that useful um you know for instance there's distinctions that people make between right what we're talking about right like whether a particular content particular represented content in the brain and if people report um experiencing it versus if people report not experiencing it right um but then there's also some people talk about levels of consciousness right which is a totally different thing you know like that's like being in a coma versus being in like a minimal you know like a vegetative state versus um you know being like awake but really sleepy you know versus being really alert right that kind of thing um you know and there's which is a totally separate thing right i mean you can kind of think of one as a as a precondition in some sense for another right like you have to be in a level of that corresponds to being awake right to have any you know or organ or like dreaming right but your brain has to be in a certain general state of of consciousness um level wise to be able to have the experience of one particular content over another right so that's can saying being awake is a precondition for studying the thing that we're interested in which is why you become conscious of one thing versus another given very similar stimulus conditions exactly you know so so there are i mean there's there's more right different distinctions right like i mean like i said there's this you know distinction between or or you know potential distinction between phenomenal consciousness right just like the what it's like to be experiencing something versus this access this thing which is roughly being being aware of the thing you're experiencing um and uh and like i said several people think this kind of thing can come apart um i also i'm very skeptical about that um um you know eg the the bridge exact the fridge flight example um chris mentioned earlier um you know i mean like i said at the end of at the end of this paper we do in some sense describe how our um can can identify something like something like two different types of access where one type of access um is kind of what's minimally necessary for empirically verifiable phenomenology um um whereas the other is more kind of what determines the actual reporting um but you know so there is something like a distinction in our model but it's not the standard kind of phenomenal consciousness versus that just um but yeah but anyway i mean there's uh my point is there's all these different sorts of things you know you mentioned like uh somebody mentioned like the like you know phenomenal like literature where people just kind of describe you know the way that phenomenology is or kind of the dynamics and phenomenology and things like that and then also as a kind of completely different very kind of descriptive you know field right that doesn't have to do with this just thinking about what becomes you know experienced versus not um so i mean the the general kind of i mean against is already kind of like long-winded answer but

just just uh i think very precise terms and very precise mappings between terms in different fields or necessary to do this kind of thing um well at all and that's not always the case all right yeah just to briefly follow up on that like i really what i'm interested is this is kind of was put out in bernard bars he called it the goodness i might butcher this i've gotten basically a minimal contrast approach where you have some content of which you're aware some content which you're unaware and up to all about up to some limit the the physical properties of the stimulus should be almost identical there might be 10 milliseconds extra between the mask and the stimulus and that's the only thing that renders it so then the idea there is that we can contrast the neuronal kind of consequences of being aware of a stimulus versus unaware independent of hopefully the physical properties of the stimulus and so so just to give an example and just to make sure that i'm being charitable um so there was recently a uh special issue of i'm fairly sure especially issue of the journal of consciousness studies on like uh consciousness in plants um and i guess one thing that i'm not entirely convinced of is that we're really talking about the same thing i think that's great and people smart we should let smart people do what they want and study all of these things but i'm not convinced for example that the process or it will take a lot of convincing to convince me that something like phenomenal conscious whatever you want to call phenomenal consciousness in a tree or a plant really has anything at all to do with this minimal contrast approach in like the neuroscience of yeah one one other one other thing again just like things kind of popping up in my head here um uh okay thanks ryan also maybe you could talk like a little closer into your microphone it gets a little quiet okay thank you all right continue and i'm really curious to see how active inference is going to play into this oh sorry sorry i just remembered what i was going to say so i think another thing that sometimes gets plated is is just um representing things versus being aware of them um right so so you know like a major so it's a for example right i mean in the in the act of the inference literature there are other papers that have described um you know things like jacobian monism right is something that like i'm like one of carl's recent papers um you know where where more or less it's a story about via the free energy principle you can come to have systems where the internal states of the system um come to parameterize or track in some way the states of the world outside of the system right so so in a really broad sense you have some some um some way in which the internal states are representing or keeping track right the external states um that as a kind of representation um or the kind of seeds of internal representations of what's kind of going on out in the world but the idea you know with this sort of literature is that the brain can represent a bunch of stuff

at the same time but only some of the stuff that's being represented becomes conscious at any given moment right so like i can flash like a word i can flash like so unlike some studies for instance i could flash the word guilt for you really fast um and you won't report ever seen the word guilt but you'll you'll behave in ways that are more consistent with uh feeling guilty or something like that right or or i mean simpler like perceptual or semantic priming examples where you might flash something really quickly again a person doesn't um you know say they are aware of having seen anything but they'll say complete word stems in one particular way versus another right if i flash an elephant for you really fast you don't know i see it you don't know you see it and i give you like a words then like el you know and it's like complete the word people are more likely to say elephant than they would be if you didn't flash it right or something like right so behavior is affected by something that was clearly caught right by the brain but but despite that it was represented um you didn't become aware of it right so i just want to make that distinction as well but being conscious of something is distinct from whether it whether the brain actually uh in some sense knew what's there yeah yeah okay so um just the idea so we used a passage over markov decision process as our model a hierarchical one i should say so this is just a single level markov decision process and just very briefly if pomdps i'm just going to call them mdps just for brevity they describe transitions among kind of hidden unobservable variables and the sensory data they generate so you can have a hidden state of the world and the observations generated by that hidden state and the goal of active inference is to infer the states and the sequences or action sequences or policies π um from the set of observations and so i'll just kind of like step through very basically how this would work so you could just have a very simple case right here where where do this is just a graphical representation of bayesian inference we can just have a hidden state and we have a prior over that state and we have some likelihood mapping between that hidden state and some a state of the world i think i'm getting feedback from someone's computer okay try again continue christopher thanks okay thanks that's better um yeah it's funny how that was how um that messes with you we've got predictive models of when our when our speed speech timing should be at least we know that you're awake um okay so we've got yeah so this this comes out to basically if you then invert this model this is just exact bayesian inference now if we start moving through time we have to add in transition probabilities and so this just imagine this first level of model is just a markov chain it's just you have some initial state you transition into your first state then you have state transitions that govern how you how those states change over time but you also have variables but you don't ever get to see this markov

chain all you get to see is a set of observations and so the idea is these observations are generated by these hidden states and so the task of inverting these models is to infer the most likely hidden state conditioned on some stimulus u_m and so you get this this is this little sigma here is a softmax function and basically you can derive message passing algorithms or using message passing algorithms rather you can derive very simple update rules that allow you to come up with posterior probabilities over states and then actions under this framework are then just kind of state transitions that the agent has control over so you can imagine in like a psychophysics experiment the actions this is a super limited exp set of action space for the agent basically they can hit left or right and they might be able to move their eyes so those would be the policies that they can control everything else the state the screen those are hidden states but they're just not transitions transitions between those hidden states aren't things that the agent can control okay so then in a way so how do we select actions well we select actions according to the free energy functional so this has two components and you select the action or the policy rather that is kind of best to expected free energy where this has two terms the first is the expected cost so this is kind of the Kullback-Leibler divergence or the difference between our predicted outcomes which is this q here and our under predicted outcomes under some policy u_m so what the outcome i expect given that i take a certain action and the predict and just our u_m and our preferred observations so these are things like having preferences for winning versus losing u_m where the agent generally is but you can encode the agent wants to win or not to lose or something like that the second term here this is the exploratory component this is the expected ambiguity where this is basically the expected entropy of the likelihood mapping it's the idea here is that you seek out actions or seek out states of the world that have a precise mapping to sensory outcomes and so for example if you're in a hotel room u_m that you've never been in before it's really dark the best way best actions to take to minimize your uncertainty is just to turn the light on u_m because that will give you a really precise mapping between the hidden states of the world out there and kind of the observations that you generate all that they generate and so the idea is that expected free energy is minimized when both of these terms are minimized and it's actually a little bit more complicated than this i should say so policy selection isn't just u_m the minimization of expected free energy it also involves minimization of variational free energy but i would just say what's the model stream u_m to get into that for our purposes in this conversation we can just say policies are selected that best minimize expected free energy and variational free energy where variational free energy is also basically a stand-in where the posterior optimal posterior

here that you infer will also kind of you minimize you you infer the optimal posterior by minimizing variational free energy and okay so what about i haven't really talked about the brain so far um one of the beautiful things back of inference is that comes the process um and so belief updates those changes to your posterior states correspond to changes in activation levels in response to input and the matrix entries here d, b, h, c blah blah blah these are all h actually isn't really a matrix i guess it's a matrix entry sure what is h entropy um so you can just these correspond to synaptic connection strengths i mean this is just like a very standard assumption across neural network literature really and just in your neural science in general where synaptic changes in weights of synaptic connections uh correspond to some kind of like function approximation okay and so then you get these kind of somewhat intimidating looking equations we can just select the ones that are really relevant so this first one here is state prediction error and so minimizing state prediction area in perception corresponds to where the corresponds to essentially the inference of the posterior probability q of s whoops and you can cast this as a gradient descent on variational free energy and that's actually where we get our erps from um christopher could you go into a little detail what the variational means independently or as related to the free energy here yeah sure so variational free energy is just to distinguish it from something like thermodynamic free energy um so like there are formal relationships between these things i would just say read carls kofristen's work if you want to know about that but basically uh variational free energy is the kullback divergence between a generative model and your approximate posterior distribution and what you're trying to do is change the approximate posterior distribution uh it slowly but surely so change it bit by bit you nudge it in the direction of steepest descent until it basically this error term here which is basically the difference between the log difference between your generative model and your approximate posterior you you change s until that difference is at the minimum and the rate at which that changes so this little $\dot{v}$ here is your erp so the idea and i'm going to go into that a little bit more in a moment um so the idea is that you have our the log probability of our posterior is essentially a non-normalized log probability it can be positive or negative like a voltage and then you have you can have your erps and this is just like your change in membrane potential over neuronal populations and that corresponds to roughly speaking like these big potentials we see at the scalp and then we have a normalized firing rate where um you will basically we normalize the depolarization variable through by branding it through a softmax function and the softmax function is just a generalization of a logistic function to more than one input more than one class more than two classes and what's nice about using a

generalization of a sigmoid function is that everyone who's looked at like rate models of pop in popular population nodes for example or somewhat sometimes called main field models um with these basically you can have this firing rate versus input curve where on the x-axis you have the amount of input that you're driving into the population and on the y-axis you'll have the firing rate of the population and it's logistic function so it's going to be bounded a roughly speaking between 0 and 1. so what you'll end up with is a this kind of sigmoid style looking curve that we treat as a probability distribution but can also be interp has kind of this lovely biophysically plausible well i don't know about plausible but biophysically sensible maybe interpretation as a normalized firing rate and that kind of allows us to translate between this kind of normative models thinking about probability distributions and what the brain is actually doing okay is everyone kind of happy with that yes stephen do you have a quick question just one quick question just is the expected free energy um on policies more important at level two and level one it kind of can soft max itself um or is it that's a bit simplifying but does does expected free energy go all the way down or is it a bit more you can generally have models that uh you can have policy selection at both levels you can policy selection at just the lower level or just the higher level depends what you're trying to model really generally speaking in models so i in the model we're going to be using i'm going to introduce in just a moment we don't have policy selection at the first level um we had policy selection at the second level but you might for example there's work by thomas parr and uh ryan and i kind of working on some similar work um as a part of my thesis where basically the idea is that the higher higher like working memory-ish pre-frontal level you have goal-directed policy selection and at the first level you just have epistemically guided policy selection um and so the policies that are being selected at the first level might be something like where in the visual field to point your attention or to move your eyes and that's just kind of determined by the salience of various items which in this framework corresponds to essentially the information gain offered by all the precision at the first level a matrix um so hopefully that answers your question yeah thanks so as a as just a little bit of kind of a generalization of that i mean really and you know and of course i'll go into this but but the way these are this is modeled officially is just each lower level trial is just a full trial as in if you were modeling it separately the higher level is just putting a prior on whatever the starting whatever the starting prior is for each lower level trial um so i mean in a sense you're modeling these try you could be you could think about each trial either at the lower level of the higher level as being modeled independently in some sense right each one can have a set of policies each can have

anything that a single level model could have it's just that the higher level is putting fires of some kind on the lower level and then the posterior observations at the lower level are then feeding back in as the observations for the higher level um so so it's just think about them as as just separate kind of trials on different time scales where they're just connected by their priors and posteriors yeah exactly um and so i guess what we did here was we wanted to have something like to implement something global neural workspace like in a deep active inference architecture and i what i want to highlight i have i have mixed feelings about defining the term the predictive global neural workspace because people lump lump this model in in a the few people who have been kind enough to cite this have lumped us in into like models of this as being kind of just another version of the globally on the workspace um i don't think that's accurate uh we took the global narrative workspace kind of and all of the phenomena that that it encompasses kind of as something that we want to explain um both ryan and i really like the global neural workspace like standard harness bars huge heroes of mine and aside from kind of like my personal love of the theory it's also among kind of neuroscientists and philosophers who work on consciousness about 30 the majority of them which is about 35 or something like that in 2018 said that they regarded as the most promising neuroscientific theory of consciousness and so we thought okay if we're going to go into this literature this is what we should be modeling i think the phenomena encompassed by this theory is where it should be the starting point but with that said it's our model is an active inference model through and through um and so it really does kind of stand on its own independently of the global around the workspace and so i just kind of wanted to flag that and so i i don't use those words or in the draft of the paper that i'm writing at the moment kind of following up on this one i don't use those words and i don't do it that's very deliberate um but anyway sorry that was an aside um that's a very esoterical side of that anyway um so the idea here is that we implement something global in the workspace like in a deep deep temporal architecture so this second level we identify deep level we identify with the workspace and so as ryan was just talking about you have these states the second level which are really like beliefs about how trials at the first level evolve over and so they provide a prior at each time step of the lower level they provide a prior over the hidden state and then after when there's posterior inference at the first level after that's happened it then shoots back up to the second and that acts as an observation for the okay so we the functions we tip the idea here is that the functions we typically associate with conscious processing take place at the level of the hierarchy that's kind of deep and temporally deep enough to abstract away from processes that

are entrained by the moment-to-moment sensory flux and this allows the agent to kind of construct a report of its experience or to do anything any other kind of kind of quite abstract cognitive task and the idea is that ignition occurs when a first level state is in third by the second level so ignition sorry is basically like what dehan has called this non-linear bifurcation of this fronto parietal network in the neuronal activity so the idea is that ignition for our model occurs when the first level state is inferred by the with a high enough posterior probability to kind of influence policy selection at the second level oops so okay on to the task so this is a very kind of general task structure which we borrowed from a really awesome empirical study done by michael pitts in 2014 and colleagues um so idea was at the first time step there was a forward mask presented and so second time step there's a target stimulus so these little lines in the center can rearrange themselves into a square and the second type there's a backwards mask of the third time step and that the agent is then asked to construct a report of its experience so whether it say whether it's or something or not um and the idea roughly speaking these little colored circles on the outside this is how we manipulate attention or and how michael pits manipulate attention um and so they either had people attend to the inside and perform a task basically click a button whenever the circle change the square changed or click a button whenever the circle changed um and so they manipulated kind of what was task relevant and i'll get into how they manipulate awareness in a but roughly speaking the way we implemented the model this is kind of a bayes net representation of the model and then if we just kind of click through it at time one there is a forward mask presented and this just corresponds to a black with a square trials that's the full this is kind of a belief at the second level there is a full sequence this is beliefs about the full sequence over the full trial so this is a time step one um there's a report hidden state factor here it's in weight because we haven't asked it to construct a report yet then at the first level each of these second level states kind of implies a first level state that makes sense and each of these first level states kind of implies a observation and attention in our model was basically we just hard coded in as a but we didn't have to hard code it you can get these things to emerge we just it's just a little bit easier in our case to do it this way we basically hard coded in so that attention corresponded to modulations of the mapping of the precision of the mapping between these first level hidden states kind of whether there was a square or lines out there in the world and the observations so okay so first first time point forwards mask we see a sequence type is going to be a square there's a trial phase one report okay next one we see trial is moved to two we've now seen the target has now rearranged itself into a square we did this for two time steps and we did the reason we did

it for represent the target stimulus the target stimulus the two time steps is to allow recurrent feedback processes from the first level kind of first level second level to kind of recognize what's going on then feed back on to the first level kind of give that's that was kind of our way of modeling this ignition style process then fourth time step we take away the math we replace the little square of just the mask again and then from five to eight do we allow the model to construct a report of its experience so basically this scene hidden state here implies a sequence of language processing states at the first i submit a square yeah just it says i see a square and then what would it say if it was unseen uh i didn't say anything i mean to be fit just to be clear like we could have and the model would be exactly the same in all aspects we could just chop off this kind of language processing part of the and the model would behave exactly the same way and we could still use it to generate reports and all of that stuff what we wanted to do was really just kind of use this as kind of an intuition pump for what having this deep temporal representation allows models to do so holding something at that level allows sequences or a cascade of hidden to kind of evolve at lower levels that might be something like constructing a sentence with a goal in mind or it might be something like um i don't know make a cup of coffee where you have to hold multiple tasks in mind and break things down the sub components blah blah blah yep as opposed to mapping to button that says yes or no if it's just a two button option then it's a one time step paradigm yeah exactly i mean we could do that i mean it'd be completely identical um this is just kind of a nice way of showing what these models can do really yeah i mean so this was this was partially for uh for a kind of to make a theoretical point right you could say right we're showing a two-level model here but you know the brain probably has you know i don't know how many levels but a very large number of levels right not just two um so you know you could you could kind of level the concern or question at us that like you know why is it that this second level is the one that corresponds to right as opposed to just any other level that's above some level right in the brain since there's a big hierarchy in the brain you know so the the idea here is is that you know we're appealing to a certain level of deep temporal structure right like um you can you can flash a stimulus at somebody and they can see it over a very short time scale right of like you know half a second or something really fast but actually but the sorts of cognitive processes that can integrate that together right and generate you know a set of thoughts right that corresponds to being aware of that right constructing a um yeah constructing a sentence or any of the things that chris mentioned um those are much deeper slower integrative cognitive right so the idea is that there is a level in the brain that is integrative enough and that represents things over a deep enough

temporal scale that it has the minimal resources to um integrate things together hold them in mind um generate sequences of the sequences of words that would be necessary to communicate that you saw something or not right so it's that abstract deep temporal structure that is the necessary conditions for what you need to get empirically um confirmable awareness yeah okay and just some brief kind of modeling notes about what we did so to model attention and stimulus strength we basically just altered the precision of the air first level a matrix mapping by passing it through a softmax function twice where one of the softmax temperature so when you pass something through a softmax function you have a temperature on it and the temperature decides how precise it is and we just had a different temperature for each iteration of the softmax function and so the idea was then we had an ordering basically of precision where the most precise was uh attention present stimulus strength high the second was attention um attention absence similar strength high third was you can imagine how that goes on and kind of the reason this is this sounds pretty abstract and a bit kind of cooked up and that's true to an extent but i think this does actually really correspond to so if you look at divisive normalization in vision where a tuning basically is a you will have some tuning kernel neuronal tuning curve and basically basically like how precise that tuning curve is is modulated by both the contrast of the stimulus and then the contrast the stimulus gets fed into some attentional process where that will together determine kind of the tuning that was a very kind of rough and ready explanation of um tuning curves but whatever um and our threshold for report we basically just set up the model to give have a preference for receiving correct versus incorrect feedback at the final time step and oaf behavior was basically modeled by simply decreasing the preference for being incorrect to encourage guessing um and so i think this was just kind of our first pass at doing report i don't think this is how the brain does report um we actually in our follow-up model we've got a much nicer way of doing this but i think it was it worked in the sense that it was able to kind of quite accurately capture a number of empirical findings so that's kind of good enough for me at this point but this first first model okay sorry that was a lot of preamble um we can get into the results so foundational simulation um this was just like kind of full like a very high precision stimulus was presented the model kind of with either on a square present or a square absent trial and this just kind of to test it um so great uh when there was a square present the model reported seeing it 100 of the time you see this nice kind of bump at the second level at the first level where um the firing rates for square increase when we present it goes down and chris let's just make sure that people understand and how to interpret these things so just just so

people understand so um time in these plots goes from left to and the grayscale from white to black corresponds to 100 probability and white corresponds to zero um probability and the gray so is you know 50 50. um so and these are you know cast in terms of firing rates here so darker than implies a higher firing rate um so the top plot on the left that says second level firing rates that just means it first sees the black circles on the outside so it's confident that it's either um gonna be a the sequence that has black circles in black square or the the second row the other sequence which would be black circles and just lines with no square um so basically once it sees the black circles then it knows okay this could either be a sequence where there's going to be a black square or when there's not going to be a black square so then at the and that and you can tell what the stimulus was because if you look at the first level firing rates so again the next kind of plot down the bottom [Music] row there corresponds to seeing the lines and the top one corresponds to seeing the square so basically what happens here is it starts out just seeing the lines and then at the second time step there you can see it kind of bumps up and the agent's really confident that it's seeing a square which at the second level corresponds to becoming fully confident that it's going to be the sequence of black circle and square and then the stimulus goes away at time point four which you can see at the first level and then stays again just lines for the rest of it um and then the the bottom two plots um show the actual outcomes um where again black is the agent's sort of competence in what it did and the um cyan dots correspond to the actual ground truth um so that's just showing um that the agent waited for the first three time steps and then recorded scene um and kept that scene while it generated the sequence of words at the following four time steps and then the right one just kind of shows it started out silent for the first four time steps and then um the words that it observed itself report were i see a square which is just those the dots moving down the diagonal to the right um so that's that's just just so people know how to interpret you know what these plots mean yeah exactly another clarifying point would just be that the same simulation is leading to all of these behavioral outcomes and neural correlate predictions and a bunch of other things so it's like one integrated model and the model stream is kind of where they built it up yeah exactly um okay so we kind of talked through all okay so in terms of the taxonomy here what we wanted to do was basically recreate that taxonomy that we talked about earlier so this was uh modulating signal strength from high to low and from kind and whether attention was present or absent and then looking at the corresponding firing rates and looking at like the percent seen and percent correct that kind of thing also sensing report behavior and percent correct which is false choice behavior okay so kind of couple of things to

highlight firing rate first level is enhanced kind of most strongly in the conscious quadrant that's nice that's again what we see in empirical findings um high firing rate for kind of square sequence at the second level even when reports are low this is kind of nice right because i talked about earlier you can have kind of a firing rate at the set you can have kind of activity in prefrontal regions that is generally speaking below the threshold for report um that is nevertheless oh sorry when i say below the threshold for report report frequency of reports are relatively low but you can still have a preference for um you can still have a higher or the representation of showing that there's a representation of the stimulus basically um and then the last thing to highlight is this that the highest firing rate for the kind of the square sequence at the second is at the is in the conscious quadrant which we kind of might think about in terms of ignition okay so then erps we see the real thing to highlight here is that we see a large p3 like event-related potential in the conscious quadrant again like a lot of the empirical findings in the report literature whereas first level first level erps seem to be fairly unmodulated by um by conscious access essentially which is again what we see so now to a very that was kind of a very abstract kind of taxonomy we were just manipulating factors what we wanted to hear was show take a concrete empirical result and show how we could basically use literally exactly the same model to reproduce it so this is a really clever paradigm by michael pitts i talked about before and so the idea is that you will have these kind of uh three phases to an experiment in phase one the participant is just told to attend to the dots on the outside of the screen and hit a button when they dim and what the what participant isn't told is that every now and again these uh things these lines in the middle will rearrange themselves into a square pattern then at the end of the first phase she seek like 300 trials and at the end of the first phase they are then kind of given a debrief and they asked all right so how many people saw a square pat in the middle and roughly speaking it's about 50 and this is now actually really well replicated result uh across kind of different varieties of intentional blindness roughly speaking about 50 of people don't report seeing it when it's not task relevant so that's nice so then they then all of those so then they have 50 of people who didn't report seeing it and then in fact are now aware of it and then phase two they do exactly the same tasks they're attending to the disks on the outside the outside uh so it's irrelevant um still task relevant but they're now all aware of it and the debris for the end of the second phase everyone reports seeing it and so now they've got this contrast between aware versus unaware so phase one versus phase two where the only thing that's different between them is whether you're aware of the stimulus and and and the stimulus in the middle is task relevant and what you can

see there's no p3b there's no late positive centro positive central um component in the erps then the final phase aware task the kind of the aware phase so all the participants are aware but they now make a task relevant to have them basically attend to the dot to the squares in the middle and when you see there is this big p3b very noticeable okay so just kind of some um this slide just summarizes what i just said then great so basically what we did was we just used a different inverse temperature parameter to modulate to kind of represent each state so phase one very low um temperature parameter for attention blur attention temperature parameter um that was kind of correspond and so this is you can kind of see the precision of the a matrix here we bump up for phase two we bump up the inverse temperature parameter just very slightly um this is kind of roughly speaking to chorus correspond to like a diffuse attention so not you're not particularly attending to one thing one particular feature of the stimulus you're attending just kind of diffusely to the whole screen and then the last state we basically model the tension make task relevant by making a very high or very precise temperature parameter where you basically get almost something that looks like a likelihood like a identity mapping okay and so these are the erps that we the erps and the firing rates do you get out of it and what you this is my favorite slide in the whole plot uh oh in the whole paper um what you see is basically results that look very very like shockingly similar to the empirical results especially for a model that is so incredibly simple um like ours is relatively at least in terms of biophysical detail um and so what you see is at the first time step so there's basically no erp so first phase basically no ips and so we we give the model feedback on every trial but what you should see is it's a little bit darker for the square for the um until we give it feedback and tell it no you were wrong um you did we did actually present squares um and the model changes its mind it was fairly sure that it had seen lines then the second one we bump up the attempt we bump up the tension parameter and now the model sees things and it gets or it reports seeing the stimulus so at this so i should say in phase one our model saw um the stimulus 49 of the time basically identified the empirical result was a fifty percent phase two it's sort of 99 of the time basically the same as the empirical results report phase two is 100 and then phase three it's a hundred percent of the time um and there there's little but by the way i should say what the way i generated these results is basically i ran several hundred tr i basically set up just ran a for loop over the stimulus over the code and for each condition and had it generate i think so each kind of iteration of the for loop generated 300 trials and i then averaged over maybe 10 iterations of that for loop so um each with a different random seed and so what you should see with the sorry that was a quick side the crucial thing is that in this last point there is this when

there's a very precise stimulus mapping at the first there is this abrupt change in firing rates at the second level and that abrupt change corresponds to the appearance or the emergence of the p3b where the p3 bit where remember that erps are generated they are the rate of change of posterior beliefs and so the model suddenly becomes very confident that saurus square and so you see this big erp boost in the p3 which corresponds to a very precise input mapping that comes along with something with a stimulus being task relevant and so there we've kind of got this really nice explanation of this kind of somewhat counterintuitive result about the dissociation of the p3 and okay and then the last thing i want we're going to touch on is just kind of we then decide to extend the taxonomy so basically this is exactly the same but we also now manipulated whether a prior was consistent with a valid expectation so we gave the model kind of we made it roughly speaking two times the model photos roughly speaking two times more likely to see a stimulus than uh see a square then lines um and or two times more likely to see lines than squares and then we presented it with squares and so that's kind of the consistent inconsistent prior and then the key result is that across all of this expectations boost the effect of feedback on firing rate which is nice and in terms of erps what we see is something this very specific prediction when there is so these are kind of like fairly small differences but they are differences um only when attention is present do you really see a p3 that's again in line with the empirical literature but crucially when there's a consistent prior or a valid phase to say a valid expectation you should see a reduced p3b in contrast to when there's kind of flat priors and when there is an inconsistent prior and this is actually something that has been empirically that has been confirmed recently again this was something michael pitcher's on this paper if anyone's interested i can kind of post a link to the paper and that that's really nice we didn't plan for that to happen um okay it is slightly annoying i would have liked to i would have liked to cite it in the paper but the paper actually came out before ours was accepted for publication oh sorry uh after hours was accepted rather that's what i meant to say um okay and just a very quick summary summary of the results with really what is a really kind of like surprisingly simple model we can reproduce a very wide range of canonical results um from minimal contrast paradigms including kind of fmri and erp findings so this includes we can account unconscious pfc activation we can account for the association of the p3 in conscious access um where the p3 roughly speaking reflects the velocity of conscious working memory updates i kind of like that phrase and four we have an explicit and formally defined role for prior expectations and the implementation of process theory and then we've got a number of key predictions that emerge from this so p3 is attention dependent and invert

and should its amplitude should be inversely proportional to expectation feedback from fronto parietal regions this is a very specific prediction feedback from frontotemporal regions should be enhanced unconscious trials and it should disinhibit granular layers in the relevant sensory population sensory cortex that's a very specific prediction that falls out of the process theory that that can be tested with dcm and then consistent expectation consistent expectations should disinhibit granule layers and sensory cortices while inconsistent price should inhibit them okay so any questions at this point christopher thanks for this awesome presentation do any of the panelists um maybe want to queue up a question how about you do this last slide with a bat leave on that sort of no if you want to otherwise we can return to the panel yeah perfect yep sounds good so all the panel prepare a question or a thought and then raise your hand when you're ready and give a last thought here thanks christopher okay so just like a brief aside on phenomenal consciousness so we associate that so i'll just back up a little bit sorry there might be a puzzle in that we associate global broadcast and the availability of information for report with this second template deep level our experience although it's kind of like there is a depth to it in the sense that experience is kind of smeared over time right there is an integration window that's relevant for experience it certainly isn't doesn't occur at the level of whole sequences and so there's a puzzle here so where does experience live as it were or where is the content of experience and kind of brian i've been chatting about this and this is the view is still evolving but broadly speaking i think what's right about this or what what we should say is that the first level states that what's what the content that we are conscious of are the first level states that have a high enough precision um either through kind of bottom-up precision or because there's a really precise prior that comes down from the second level a matrix um so basically the message that's being passed back up to the second level has a high enough precision to influence policy selection or to pass the threshold the posterior belief threshold for conscious access and report and so this kind of has the nice consequence that it keeps experience at the relevant so it's updated at each time step when we get new stimulus in stimuli in and yet it has a distinct link to report and behavior yeah so i mean just to expand on expanding that a little bit i mean the kind of idea is just that as opposed to and again chris partially said this but just kind of reiterate um the idea is that um what you consciously experience um is is a consequence of both the posterior states at the first level and whatever the second level likelihood is right so it's a function of both the posterior there s_1 and what the structure is of a_2 there and you know way to think about that is if you were to rearrange a_2 so the mapping was different um and s_1 had the same posterior then that would update the beliefs

at level two in a different way right so you know if i arrange a2 one even though s1 there is still sensitive to the same stimulus um then if i rearrange a2 one way then the posterior at s1 is going to lead to um awareness of one thing whereas if a2 is set up a different way then the second level will gain awareness of a different thing right so so in some way the phenomenal content that you become aware of depends on the nature of this link between what messages the way that the messages at the second level are kind of decoded in a sense right what the second level takes to be the meaning of the messages that get passed off from the first level um so so this kind of nice thing like chris mentioned where it's the updates um that the first level gives to the second level at each time point so phenomenology kind of stays at the temporal scale of experience but uh phenomenal content still depends on a certain type of access right it depends on the first level having a specific type of influence on the second level brian thanks a lot it's really interesting so let's return to the guest the panel mode christopher awesome presentation and the first question is going to be from dave and then anyone else can raise their hand so dave go ahead my question is about possible involvement of the air toss the reticulothalamic activating system either in your data or the data that are uh drawn on neurological data in the other relevant studies is there any representation of possible contributions by the air toss by the midbrain and is there any accommodation in your of something maybe not explicit with the air trust but uh data uh that might originate there say the endogenous um so subcortical structures are really interesting and as for the reticular thymic activating system i'm curious did you did you get that from this book by bernard barz so a cognitive theory of consciousness where he first introduced global workspace theory and he has a whole chapter devoted to the reticulophalamic activating system oh that's great i will definitely have to get that no actually i got that this from uh the that uh yak punk sepp and uh mark solms did especially the 2013 thesis of the the conscious id he is of course a very close collaborator with carl frist and he just brought out a full-length book on the conscious id especially as um active inference interacts with that so so my my actual main focus of research is on um emotions um you know so applying computational models to emotions and um i'll just say that i am i have very strong disagreements with the um the kind of theory proposed by jack and mark we actually published kind of a joint like on paper debate where me and um a colleague richard lane debated back and forth with jack and mark about the right way to think of um the way consciousness works and i'll just say that you know their view places this really strong role of affect where they they more or less say that these midbrain you know structures um like the pag right the um gray um send these afferent signals up to kind of activate the rest

of the brain and that somehow permeates it with this kind of necessary emotion um and that's kind of primary um whereas whereas kind of the way that um the way that we've the kind of structure of the sort of thing that we've defended is just that no like the global broadcasting ish kind of thing applies to all types of information right so if you have a representation of your beliefs about your emotional state that has to be attended to and sort of sent into the workspace just like any other stimulus for you to become aware of it um and so i mean these sorts of midbrain structures they do a lot of really interesting like interreceptive and visceral regulation processes um and you know they do have this sort of general like upward modulation rule while they well they kind of you know my view is they kind of keep the cortex in general in a state where it's capable of representing things being sensitive to simulate holding different states active and other states not so it essentially it allows things to be represented in general and then to become consciously accessible um but you know with respect to um the way that kind of emotions feed in and whether the uh the midbrain structures actually contribute content in and of themselves we just have theoretical disagreements about that can i also briefly follow up on that so the contrast the reason i think that fronto-parietal regions in cortex are so important is because there are a huge just amount of empirical literature now in neuroimaging also very recently data from kind of invasive neurophysiology in monkeys and most recently in mice showing that yes when there is this contrast between conscious versus unconscious these are the things that differentially modulated now you can't there are some very impressive findings very recently coming from matthew larkham's group showing basically that you can wake creatures up from general anesthesia by stimulating the medial dorsal nucleus and the thalamus and there are these crucial connect like reciprocal connections between layer five of the cortex and the medial dorso and the thalamus and when those goes out basically uh general they go out under general anesthesia and when you are awake or when like whiskers are stimulated in barrel cortex when they report see when the mouse kind of reports as in it says its whiskers were stimulated this circuit goes off essentially so i think the thalamus is crucial but uh kind of allah dehaan and every everyone else i think the reticulophamic activating system and all those sort of cortical structures these are background conditions similar to what ryan said it puts the cortex in kind of the state where it could be aware of something but the actual contrast between something being broadcast or broadcast first not broadcast or being reported but not reported does not rely on those structures um and i would just want to see from those authors i i would actually just want to see them engage with the visual awareness literature and show that those structures are differentially involved thanks

for that uh steven go for it and then dean blue oh actually then scott then dean or blue if you want but stephen with a question then scott yes so thanks um just a question you you just to be clear so effectively you tagged the degree in prediction confidence and that mapped against the the data that's empirical if i'm writing understanding so you like as the model showed a difference in confidence that was showing a correlation to how firing happens and i'll just be curious to know if um you think that this this sort of key level or something where this firing or deductive piece can happen um where there's this where there's a shift between and maybe a more markovian ergodic process to something more semi-markovian um and less ergodic and you so that an organism can make non-ergodic choices in conscious uh so i mean a couple of things so i'm not sure about the ergodicity comment so i think you can probably have i'm not sure how semi-markovian systems relate to ergodicity i assume that you can have a semi-markovian system in a godix system that doesn't like that wouldn't surprise me um yeah so basically when you move from uh i just don't know actually just to flag that but if you i'm not particularly up on a lot of the physics cisco physics that relates to how these models work [Applause] but yes when you move from a first level model to a second level model they are sammy markovian and can you just maybe define that a little more what does that mean to you what does that enable and maybe that will get at what stephen was asking too um basically the model is no longer just purely dependent upon the previous state at the first level of the model um there is more information that is determined at the second level than what is just goes on at each time step of the first level that's basically it uh yeah so i think we we have a sentence about this somewhere yes uh you what what the temporally deep scale buys you is essentially what we were saying before all of those cognitive actions that allow you to do things like construct reports make plans etc etc and kind of abstract away from moment by moment sensory flux all of these things depend upon there being a second level and that might be reflected in the brain there's some from the ground yeah yeah so smooth uh no it's a great grand depressor granular cortex is generally speaking like so the cortex has six layers roughly speaking across and as you kind of move up or move towards the center of some centrifugal hierarchy depending on who you talk to and what kind of anatomical maps you believe that type of thing layer four of that the granular cortex which is essentially the input layer from the thalamus and the feed forward layer that gets smaller and smaller and smaller until when you're at kind of the center of this hierarchy or this centrifuge um you essentially have no granular cortex at all uh this is kind of ties in lisafeld and barrett's done a lot of stuff with this i have problems with that view to be honest they kind of say the top of the hierarchy is where

they think consciousness lives i have a rather vehement disagreement in that i just think there's absolutely no evidence that that's true in terms of neuroimaging evidence uh the areas of the relevant seem to be these fronto parietal regions uh they do have a layer they do have a somewhat well-defined layer for it's not completely a granular it's not granular somewhere in between um yeah i don't know i think they what one thing to say about i know ryan has very specific views about this and i should say that i really love lisa from barrett's work i think she's wrong about this um i i i just want to go with kind of the empirical contrasts right and not speculate too much about very specific implementation details so look they kind of go from a very loose model of kind of predictive coding and then map it to these like cellular uh kind of details about the structure of neurons um i'm more interested in going from these computational properties to basically what would measure the level of neural imaging because i think that's what is most tractable experimentally and then if we're and there's still a lot of debates about that neuroimaging level right if we kind of hammer it out and decide what's going on at the neuro imaging level of analysis then good then then we can start having more serious discussions about particular involvement particular types of cortex but i don't think we're there yet empirically yeah i mean i i would say yeah just to kind of echo that a little bit um you know it's it's important to realize that um it's not as though you know the particular sets of you know little node neurons and synaptic um in the kind of like you know hypothetical columns um you know that chris showed earlier in terms of the neural process um that's just kind of one example of a way you could set it up there's there's a very very large number of ways that you could connect up neurons together in different structures that would implement basically the same algorithm and in addition to that there are different message passing algorithms that you can use um to solve the same graphs um an active inference isn't defined by some particular message passing algorithm right like thomas parr has shown this you know specifically in a paper where you can have different neural implementations that require more or less neuronal resources to implement it's kind of a trade-off between efficiency and accuracy where you know for instance like really really accurate um message passing algorithms like belief propagation they require more neurons to do um but they're a little more accurate and the brain could use that right or the brain could use variational message passing which is not quite as good of an approximation but requires less resources to do um so the point being is is that it's not really that strong of commitment to that level of kind of circuit detail in active inference it's more just a theory of you can have this pattern of synaptic connections that implement the matrices and we represent posterior beliefs over states

with these firing rate functions um and then you take the rate of change in those to put the brps um but so this is this is kind of the the issue with it doesn't really make sense to me anyway at this point against echoing with what said um to start with hey let's assume that this very specific way of setting up neurons to do this is the right one you know let's just assume that and then figure out what must be true about different areas of cortex based on their layer based on their cortical column layer structure um so so then it's definitely i mean it's super interesting and probably has um you know a really meaningful interesting mapping between the differences between granular and granular and a granular cortices um with respect to computational function um but um but again you know like i don't think um until we've really nailed down that there is this one particular circuit implementation um it doesn't really make sense um in my opinion anyway to kind of infer from that direction um to function um so i should say yeah just to kind of echo that i think if you work in your imaging it doesn't make sense if you worked in mouse as a in a mouse uh model and you were literally testing hypotheses at the circuit level then it would make sense something to do but well right but in that case but in that case you're literally testing like the predictions of specific message passing algorithms yeah exactly so that was kind of testing that's testing for message passing algorithms that's not assuming a particular one and inferring from that so anyway i mean but i mean if you i mean for people who are interested i mean i mean we you know so so i and some pre a couple previous papers have um you know kind of built um multi-level active inference models similar to this but specifically about um emotional awareness um you know the the sort of show how you can get um you know from simple kind of um lack of awareness of single emotional states all the way up to being aware of feeling multiple emotions at the same time and being able to report them and things like that that kind of show how a structure not too different from this one can can generalize to um to emotions or really to to any other sort of thing that you could experience and be aware of um whether that's coming from inferring things about your own kind of bodily state and its emotional meaning from interception or whether it's something like vision so i mean we we haven't covered that stuff in any of these sorts of things before but but um that stuff is also out there um that might kind of help at least show what kind of the view on our end is about how active inference can uh account for um emotional phenomena and be consistent with the current literature on emotional awareness as well thanks for these awesome answers scott with a question and then i'll ask one from the chat and then dean or blue or someone else there we go um thanks great and fascinating presentation i was wondering this book the intelligent movement machine

yeah and um it's it it talks generally about the um cortical mapping being predicted by the movement repertoire in the motor cortex and kind of that embodiment kind of element one of the things i wanted to ask about the reason i think it follows from that this most immediate prior part of the conversation is when we have an active inference setting and that model is being employed for multiple um perception and i won't use the phrasing right because my active inference terms aren't all up to stuff but when the inputs into the individual who's performing the active inference that system are from various sensory inputs and various temporal distance so there's several inputs and then they're being synthesized into a an action or a an externality something's going to reach out and again my apologies for not internalizing all the definitionals um what is what are some of the cortical mappings or have you found cortical mappings that embody that process of synthesizing multiple inputs into a single action what are the core what are the structural correlates of the process of synthesis of multiple different strands of active inference into a single action potential again i'm using the wrong words but nice questions this is a question essentially about multimodal integration yes so yeah yeah um so i guess i i think active inference is probably you might be able to build a specific so i think there are two parts this question that's an interesting question first part is i'll just kind of answer the active inference end of it uh the way i think about active inference at least is just as a really useful modeling and mathematical framework and i think it's possible to build multiple active inference models of any phenomena so i would want to hammer out an empirical paradigm um and then figure out okay what models could i build of this so that's kind of the first part the second part is about multi-modal integration are there kind places that house moji mode i haven't looked at modi modal integration from since i was an undergrad i'm just trying to remember like my third year cog neuro courses i think there are places in like posterior parietal cortex where there is integration between multiple modal like v you might say like vision and one other thing or whatever it is and then that gets shot off to motor cortex implemented in actions um so i don't know i think you i would just say look at the motor planning literature um there's lots of stuff about uh bayesian um visual motor integration and one and just follow up one of the things that that leads to is i was wondering about the situated cognition opportunities for those syntheses so for instance you have um hysteresis or my son used to say stereotypes are real time saver he said it as a joke right so the idea is you have externalities that facilitate the synthesis so they're actually not cortical structures they're social and linguistic and rhetorical structures right i mean is that could those what i'm where i'm exploring is could those be described with some of the same models as the cortical structures

so you're actually able to have a scale independent description of the processes both in a social and a cortical cognitive context for the synthesis of multiple multimodal inputs so i'm not sure about the scale independent part yeah yeah i i i hear what you're you know scott it's a it's a beautiful question i see where you're coming from because we've been talking about the scale-free formalisms with crisp fields and others and then this whole nested markov blankets like you could have the neuron and then the neural you could go within a model and just like you said um your model doesn't presuppose that awareness is actually at a higher level you're just saying there is a structure that's deep temporal structure and that's the state that's the timeline that you are at but there could be another level and then mechanistically or functionally how that level plays out is going to come back to a task specific measurement which is and a plurality of so still good points i mean part of part of i mean i think you know timing to go back to i mean there's lots of like nice opportunities right to build active inference models would be very simple you know for like multiple multi-modal or a cross modal right integration task you know like for instance like um i haven't seen this done um but maybe i have to look back but things like the double flash illusion or the rubber hand illusion or you know any any of these kinds of yeah multi-modal things i mean like the like the double flash illusion is really nice right because i mean more or less for people who are familiar um you either play one or two tones and they are coincident with a single flash of light um and it just turns out that when you play two tones really quick then people perceived two flashes of light when there was really only one um and the reason for that is so there's just this kind of like friar in the mix presumably that coincident you know things that are coincident temporarily have the same hidden cause but also at the same time um the temporal resolution of audition is just better um this is more trustable than the temporal resolution of vision um so so you can just say look like the system just trusts right it has beliefs that the auditory system has higher precision and that there's a high probability of common causes of coincident input therefore there must have been two flashes um given that there was two tones um you know i mean that's just one example of like a really simple well-known empirical task of cross-modal integration that would be pretty easy i mean probably without a lot of without even a ton of tweaks to the kind of structure of the model um that we're using here you could um do all you'd really have to do probably is just assign the right precisions right to the auditory and visual input where those are jointly um generated by the same um hidden cause or hidden state in the um and so basically it's either you're just inferring one cause two causes and that either generates receiving one flash or two flashes um i mean anyway it would be it would be simple

to do um and uh i guess that i don't know if i don't not aware of that having been done in the kind of formal active inference mdp literature but um you know i mean there are known right like neural correlates of this stuff and you know they're in part where you'd expect them to be right they're at like the borders right of like auditory and visual processing and like association cortex um like posterior association cortex um and uh you know it'd be really really really cool project right to try to build a model of something like that and just see right whether the predicted uh neural time courses um actually better kind of single out right the the areas uh the association areas where the kind of borders division it seems like it's it's almost like a situated synesthesia where you have the gabor square where you have the frequency and time variance but it's being one of the variants is being hijacked by another perception in the visual field right you know so your your it that you it's crossing over to that other sense it's very interesting thank you thanks um i'm gonna ask a question from the chat and then if anyone wants to give probably a short final note but then we'll end around the hour so that we can have a whole second discussion next week the question the chat is um to the authors what would you say the main limitations of the model are or where are some next things you're building on but what are the main limitations of the current model and then how are you going to be building so i mean there's lots right uh so i'll just list them i think we list this in the paper somewhere too um for starters it's a discrete model we treat the whole visual system as basically like one discrete hidden state to be inferred by the high level that's obviously like not true but it's a good enough approximation to actually think reproduce a bunch of it would be nice to have a multi-level now with that multi-level model i think there are some empirical cases there's this thing called the partial awareness hypothesis by sid cudeva where they talk about having partial access or the global workspace having access to different parts of the hierarchy and that giving rise to kind odd experiences so you might have something like you might have really good access to the fact that uh the color of ryan's shirt but not to the fact that it's a shirt or vice versa um so that would be nice i'm not sure quite how to do that in a hidden kind of hidden markov model situation um but that would be nice second thing is um we really just model kind of report paradigms here so where where we're going with this is we're trying to extend this explicitly to model some no report paradigms and actually casting kind of thinking about that in a more precise and so we've actually built another model and we're kind of writing it up at the moment um i mean it's not it's kind of a side because it's not a part of my phd thesis the idea is roughly speaking to actually have a model that is able to explain a lot of the weird findings in the no report literature and also i think we have a much more principled

we now have a much more principled account of report that i'm really quite excited about um then just like i guess on a last limitation would just be uh and this isn't really a limitation because ryan has got a lot of work in emotion and interception on this and you know that's somewhere that his lab is like really pushing um but it would just be nice if we could use these very similar architectures and apply them to something like intraceptive and i would be i would find it personally very compelling if you had one i think it's too demanding to have one model but one mod very general model structure that generalizes across visual consciousness to intrceptive awareness to like auditory to smarter sensory awareness that would be really nice awesome answer ryan do you want to add anything and then we'll have any final comments um you know i mean i think i think chris covered most of the the kinds of limitations or i mean i don't know i mean i would just say like like yes i mean it would be it would definitely be nice to do something like have a so they're what are called mixed models um where you have kind of a discreet model like ours on the top but then the um the kind of posteriors some predictions downward that essentially set set points for a continuous level below it um so so then you can have a continuous state space where where for instance you know it's a lot more plausible right to think that um these sorts of inferences that are being made in kind of like early visual processing or on a you know on a continuous continuous state space rate it doesn't have to be like motion speed 1 motion speed 2 right it can be it can be continuous values and for for things like that um but um like another thing that i you know i mean you know perspective interception and emotion i mean that's something like i said i mean we have empirical paradigms that we've been setting up to to test just this kind of thing in uh in interception um and we have a we have a paper under review right now where we apply a computational model to um interreceptive uh perception and like a gastrointestinal interception task that's not a multi-level model at the moment um and that paradigm doesn't have um probably doesn't have enough trials to do contrast between um conscious versus unconscious um stimulations but we're kind of working on that um so that would be nice um another that um you know chris kind of mentioned the example of like being aware that shirt is red but not a shirt right that kind of thing also is really important for modeling emotional awareness because um a lot of times right people might feel right some kind of bodily sensations you know kind of like a pit in their stomach or like fast heart rate or something like that right they might be conscious of that but not infer one step above um and be conscious of the fact that that's associated say with like feeling fear right like like uh people with like panic disorder for example right they might feel like a strong heartbeat but they might instead infer

oh that means i'm having a panic attack or something like that or they might infer oh i'm having a heart attack right so there are different kind of conceptualizations that you can you can infer from right patterns in lower level interceptive experience some of which are emotional and some of which are not um but and this is where it becomes you know interesting is that um you might also prime right an emotion category right like like i mentioned with the guilt example right i could prime guilt so that higher level concept representation um is activated but also but does not enter into awareness right so you can kind of have multiple levels in a where each level can independently or compete for broadcasting um you know so you know i've been thinking about ways to do this model wise and the the best thing we've come up with right now just in terms of the practical implementations we have would be to for instance like have an intercepted level um and then have the emotional level above that but have the interception representations also get kind of like duplicated and passed up to the second level so then they can also kind of equally compete right for access to like a third level that would be like the the workspace um but anyways i mean these are all things that are kind of in progress but just by way of saying um this additional limitation is you can only be in our model you could only be conscious and of conscious or unconscious of one type of processing or have one level of abstraction um which not you know the case in the which is not the case in the it's not the true case thanks a lot for these awesome answers always a great time with the two of you on the stream so just thank all the participants who joined us live and in the calendar event there's a forum so be helpful if you want to add feedback on the form and then we're going to be having 18.2 next week at the same time it's going to be the same paper so hopefully we'll all get another chance to reread the paper listen to this stream because christopher what you said was really deep so again much appreciated to all and we'll see everyone next week thanks guys